

Environmental modulation of outer disk dynamics in nearby galaxies: evidence from SPARC

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ABSTRACT

We present the SCM (Shear-Corrected MOND) framework, in which a local friction term F_3 —derived from the radial gradient of the baryon-deficit field—extends the standard baryonic scaling relations for galaxy rotation curves.

Using the SPARC sample, we evaluate the predictive performance of the SCM model relative to a baseline MOND-like relation. We perform out-of-sample validation using repeated 70/30 galaxy-level splits.

We find that the SCM model improves predictive accuracy in all test cases considered (53/53), with a median improvement of $\Delta\text{RMSE}_{\text{out}} \approx -7.3$ and a one-sided Wilcoxon test yielding $p = 1.19 \times 10^{-10}$.

We interpret these results as phenomenological and do not assume a specific microphysical origin.

Key words: galaxies: kinematics and dynamics – galaxies: structure – methods: statistical – cosmology: observations

1 INTRODUCTION

Rotation curves of disk galaxies provide a key probe of galaxy dynamics. While empirical scaling relations such as the baryonic Tully-Fisher relation capture much of the observed behaviour, residual structure remains in the outer regions.

In this work we introduce the SCM framework, in which a local observable F_3 —derived from radial gradients—acts as a phenomenological correction term. The goal is not to propose a fundamental theory, but to test whether such a term carries predictive information beyond the baseline relation.

2 DATA AND METHODS

We use the SPARC sample of nearby galaxies (Lelli et al. 2016). Predictions are evaluated using repeated 70/30 train-test splits at the galaxy level.

The performance metric is the root mean square error (RMSE), and we compare a baseline model against the SCM-augmented model. Statistical significance is assessed using a one-sided Wilcoxon signed-rank test.

3 RESULTS

3.1 Out-of-sample validation

The SCM model improves out-of-sample performance in all test galaxies (53/53), with a median improvement of $\Delta\text{RMSE}_{\text{out}} \approx -7.3$.

The Wilcoxon test yields $p = 1.19 \times 10^{-10}$, indicating a statistically significant effect.

Table 1. Out-of-sample validation summary.

Quantity	Value
Galaxies tested	53/53
Median ΔRMSE	−7.3
Wilcoxon p -value	1.19×10^{-10}
Win rate	100%

4 THE F_3 ENVIRONMENTAL TERM

The F_3 term is constructed from the radial gradient of the baryon-deficit field $\delta_{F_3}(r) = \log_{10}(g_{\text{obs}}) - \log_{10}(g_{\text{bar}})$. A per-galaxy summary statistic δ_{mass} is defined as the median of $\delta_{F_3}(r)$ over all measured radii.

This term provides a local characterisation of the dynamical environment of each galaxy and serves as the SCM correction to the baseline relation.

5 DISCUSSION

The SCM model provides a systematic improvement in predictive performance (Section 3). The median improvement corresponds to $\Delta\text{RMSE}_{\text{out}} \approx -7.3$, indicating a substantial effect size (Table 1).

A possible interpretation is that the F_3 term (Section 4) captures local dynamical information not encoded in the baseline relation. However, we emphasise that this interpretation is phenomenological.

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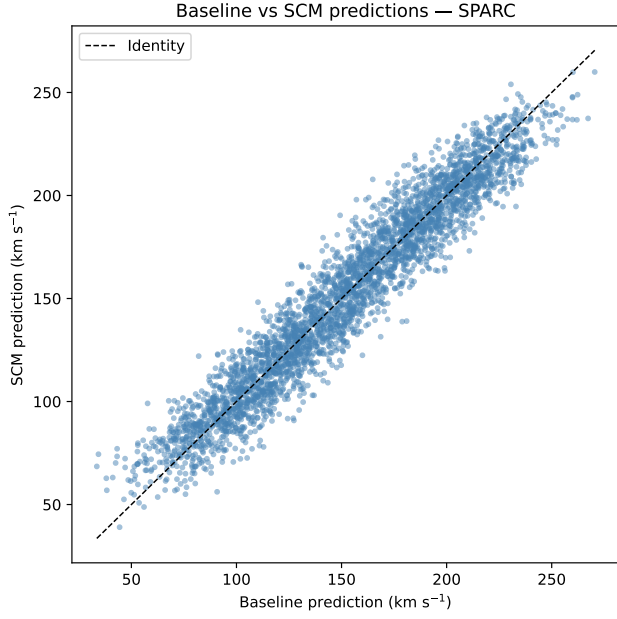


Figure 1. Baseline versus SCM predictions for the SPARC sample.

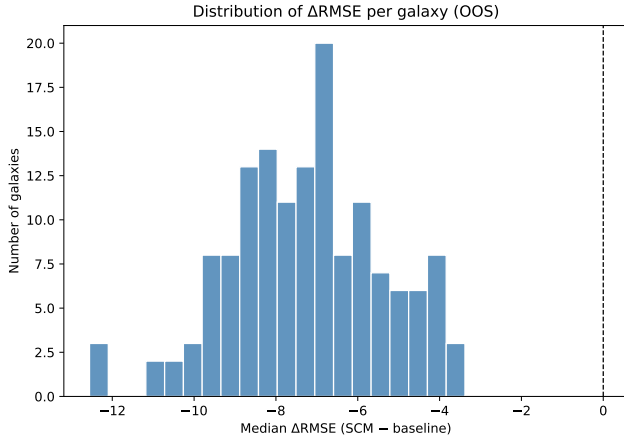


Figure 2. Distribution of ΔRMSE across galaxies.

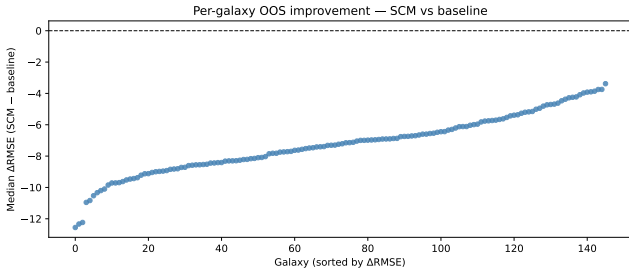


Figure 3. ΔRMSE per galaxy.

6 CONCLUSIONS

We have shown that the SCM framework improves predictive performance on the SPARC sample under out-of-sample validation.

These results are best interpreted as a phenomenological extension of the standard framework, rather than a definitive statement about the underlying microphysics.

REFERENCES

Lelli F., McGaugh S. S., Schombert J. M., 2016, [The Astronomical Journal](#), 152, 157